Chapter 3
Vector Spaces

3.2 Subspace, null space, column space and row space

Learning outcomes

- Be able to show whether a subset is a subspace or not
  判断向量空间的子集是不是“子空间”，要检查加法和标量乘法是否闭合
- Be familiar with examples of subspaces:
  - Null space of a matrix “齐次方程组的解集”
  - Span of vectors “所有线性组合”
  - Column space “列向量的所有线性组合”
  - Row space “行向量的线性组合”
- Be able to check whether a set of vectors is a spanning set of $\mathbb{R}^{n}$, by row reduction, or by finding a vector not in the span.

Subspaces of Vector Spaces

Definition

Let V be a vector space (for example, $\mathbb{R}^n$). A nonempty subset S of V is a **subspace** if it is closed under addition and closed under scalar multiplication.

In other words,

(1) $\mathbf{u}, \mathbf{v} \in S \Rightarrow \mathbf{u}+\mathbf{v} \in S$ (*S* is closed under addition)

(2) $\mathbf{u} \in S, c \in \mathbf{R} \Rightarrow c\mathbf{u} \in S$ (*S* is closed under scalar multiplication)

To show that a subset S of a vector space forms a subspace, we must show that

1. S is nonempty (it contains the zero vector)
2. The closure properties (i) and (ii) are satisfied.

Example 1

Consider the subset W of $\mathbb{R}^2$ of vectors of the form $[a, 2a]^T$
Show that W is a subspace of $\mathbb{R}^2$.

Proof

Observe: W is the set of vectors that can be written as $a[1,2]^T$

- W is non-empty because $[0,0]^T \in W$

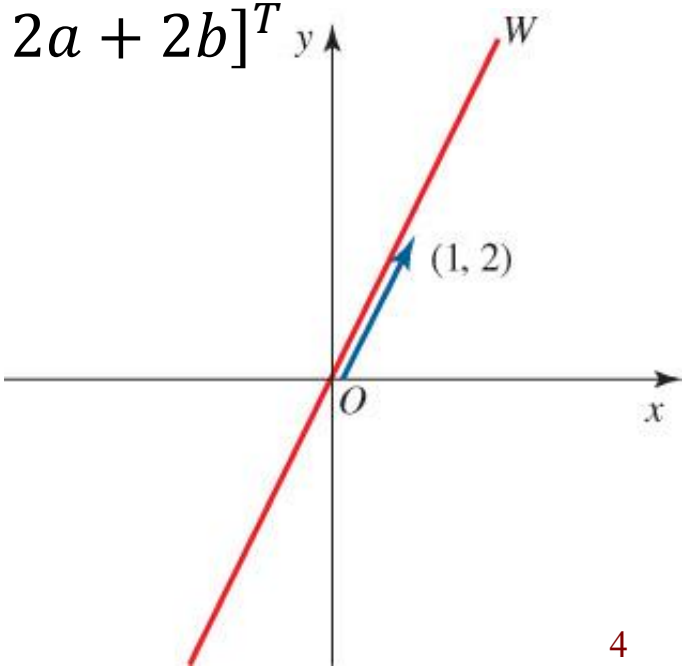
Let **u** = $[a, 2a]^T$, **v** = $[b, 2b]^T \in W$, and $k \in$ **R**.

- **u** + **v** = $[a, 2a]^T + [b, 2b]^T = [a + b, 2a + 2b]^T$
  $= [a + b, 2(a + b)]^T \in W$,
- k**u** = $k[a, 2a]^T = [ka, 2ka]^T \in W$.

Thus **u** + **v** $\in W$ and k**u** $\in W$.

W is closed under addition and scalar multiplication.

Therefore W is a subspace of $\mathbb{R}^2$.



Example 2

Consider the homogeneous system of linear equations. It can be shown that there are many solutions $x_1=2r$, $x_2=5r$, $x_3=r$. We can write these solutions as vectors in $\mathbb{R}^3$ as $(2r, 5r, r)$.

$$\begin{aligned} x_1 - x_2 + 3x_3 &= 0 \\ x_2 - 5x_3 &= 0 \\ 2x_1 - x_2 + x_3 &= 0 \end{aligned}$$

Show that the set of solutions W is a subspace of $\mathbb{R}^3$.

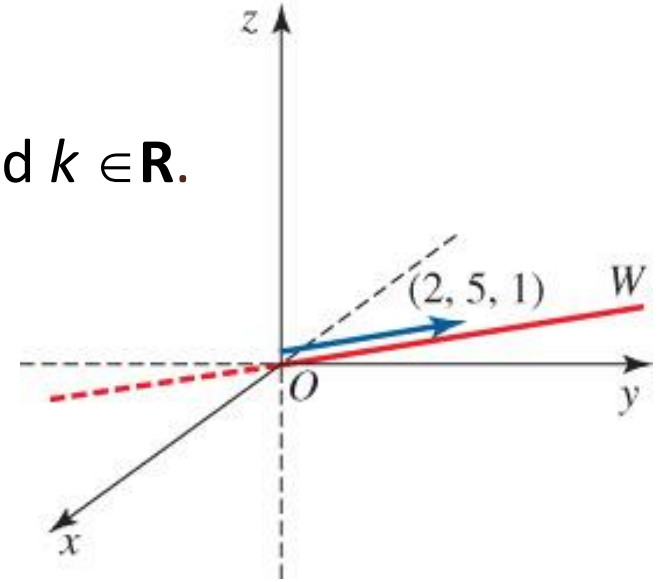
Proof

- W is nonempty: $[0,0,0]^T \in W$

Let $\mathbf{u} = [2r,5r,r]^T$, $\mathbf{v} = [2s,5s,s]^T \in W$, and $k \in \mathbf{R}$.

- $\mathbf{u} + \mathbf{v} = [2(r+s), 5(r+s), r+s]^T \in W$
- $k\mathbf{u} = [2kr, 5kr, kr]^T \in W$

Thus $\mathbf{u} + \mathbf{v} \in W$ and $k\mathbf{u} \in W$,
and W is a subspace of $\mathbb{R}^3$.



Observe: W is the set of vectors that can be written $r(2,5,1)$.

Exercise

Let $S = \{A \in \mathbb{R}^{2\times 2} : a_{11} = 2a_{22}, a_{21} = 0\}$ with standard addition and scalar multiplication. Prove that S is a subspace of $\mathbb{R}^{2\times 2}$.

Poof: Step 1: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in S$, therefore S is nonempty.

Step 2: Let $A = \begin{bmatrix} 2a & * \\ 0 & a \end{bmatrix}$ and $B = \begin{bmatrix} 2b & * \\ 0 & b \end{bmatrix}$ be any 2 elements in S.

$$A + B = \begin{bmatrix} 2(a+b) & * \\ 0 & a+b \end{bmatrix} \in S$$

So S is closed under addition.

For any $k \in R$,

$$kA = \begin{bmatrix} 2ka & * \\ 0 & ka \end{bmatrix} \in S$$

Therefore S is closed under scalar multiplication.

Therefore S is a subspace of $R^{2\times 2}$.

Trivial subspace

If V is a vector space, then V is a subspace of itself, and S = {**0**} is a **subspace** of V.

All the other subspaces are referred to as ***proper subspaces***.

We refer to {**0**} as the ***zero subspace*.**

Null space of a Matrix:

For an $m \times n$ matrix A, let $N(A)$ denote the set of all solutions to the homogeneous system $A\mathbf{x} = 0$, that is

$$N(A) = \{x \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

$N(A)$ is a **subspace** of $\mathbb{R}^n$ because:

- **Closed under addition**:

  if $\boldsymbol{x}, \boldsymbol{y} \in N(A)$, then $A\boldsymbol{x} = \mathbf{0},\ A\boldsymbol{y} = \mathbf{0} \Rightarrow A(\boldsymbol{x} + \boldsymbol{y}) = \mathbf{0}$, so $x + y \in N(A)$

- **Closed under scalar multiplication:**

  if $\boldsymbol{x} \in N(A), \mathrm{c} \in \mathbb{R}$, then $A\boldsymbol{x} = \mathbf{0} \Rightarrow A(c\boldsymbol{x}) = \mathbf{0} \Rightarrow c\boldsymbol{x} \in N(A)$

The subspace $N(A)$ is called the **null space** of the matrix *A*.

How to find the null space of a matrix A?

Very Important!

Example: Very important

Find the null space $N(A)$ for the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$.

Example: Very important

Find the null space $N(A)$ for the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$.

Solution

Use Goss-Jordan reduction to solve $A\mathbf{x} = 0$, we obtain

$$[\boldsymbol{A}|\boldsymbol{0}] \sim \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 \end{array}\right]$$

How to determine leading variables and free variables?

The reduced echelon form involves two free variables, x_3, x_4:

$$x_1 = x_3 - x_4$$
$$x_2 = -2x_3 + x_4$$

Write leading variables in terms of free variables

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_3 - x_4 \\ -2x_3 + x_4 \\ x_3 + 0x_4 \\ 0x_3 + x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Then, $N(A) = \left\{ \alpha \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} | \alpha, \beta \in R \right\}$

Write all the variables in a column vector, and replace the leading variable by free variables

Split the vector x, in terms of linear combinations of known vectors, with free variables as coefficients.

The Span of a Set of Vectors

- Definition: For vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ from a vector space V and scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$, the sum $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n$ is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$. 线性组合
- The set of all possible linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is called the **span** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, denoted by $Span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$: 张成（的空间）

$$span(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n) = \{\alpha_1\boldsymbol{v}_1 + \alpha_2\boldsymbol{v}_2 + \cdots + \alpha_n\boldsymbol{v}_n : \alpha_i \in R\}$$

Examples:

- The span of 1 nonzero vector in 2D or 3D is a straight line 直线
- The span of 2 vectors in 3D is (usually) a plane 平面
- In previous example, the null space of A is the span of vectors $[1, -2, 1, 0]^T$ and $[-1, 1, 0, 1]^T$

Example:

Given $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$, what is span(**v**)?

In words: span(**v**) is the set of vectors of the form

$$\alpha\mathbf{v} = \alpha \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha \\ -2\alpha \\ 0 \end{bmatrix}$$

In notation:

$$span(\mathbf{v}) = \left\{ \begin{bmatrix} \alpha \\ -2\alpha \\ 0 \end{bmatrix} : \alpha \in R \right\}$$

Theorem

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are vectors from a vector space V , then $Span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is a subspace of V.

Why is it a subspace?

- If two vectors are both linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, is their sum still a linear combination of the n vectors?
- If a vector is a linear combination, is its scalar multiple still a linear combination?

Spanning Set

Definition

The set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a **spanning set** (向量空间的）张集 for a vector space V if and only if every vector in V can be written as a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, that is to say, for any $\mathbf{x} \epsilon V$ there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n.$$

如果某些向量$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$是V的张集，则V中任意向量都可以表示为$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$的线性组合

How to check if a set spans $\mathbb{R}^n$

Is $\{v_1, v_2, ...\}$ a spanning set of $\mathbb{R}^3$?

To determine whether a set spans R^3, we must determine whether an arbitrary vector $\mathbf{b}=[a, b, c]^T$ in $\mathbb{R}^3$ can be written as a linear combination of the vectors in the set.

- Step 1: Put the column vectors into a matrix, A=$[v_1, v_2, \ldots]$
- Step 2: Augment the matrix by the vector $\mathbf{b}=[a, b, c]^T$
- Step 3: Row reduce the augmented matrix [A| **b**] to check whether it is consistent for ALL a, b, c
- Step 4: If [A| **b**] is consistent for ALL a, b, c, then the columns of A, $v_1, v_2, \ldots$, is a spanning set of $\mathbb{R}^3$.

Spanning set example 1

- Is $\{e_1, e_2, e_3, (1, 2, 3)^T\}$ a spanning set of $\mathbb{R}^3$?

Spanning set example 1

- Is $\{e_1, e_2, e_3, (1, 2, 3)^T\}$ a spanning set of $\mathbb{R}^3$?

Solution:

It is easily seen that $[a, b, c]^T$ can be written as

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

So the set is a spanning set of $\mathbb{R}^3$

Spanning set example 2

- Is $\{ [1, 1, 1]^T, [1, 1, 0]^T, [1, 0, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

Spanning set example 2

- Is $\{ [1, 1, 1]^T, [1, 1, 0]^T, [1, 0, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

Solution:

We must determine whether it is possible to find constants α_1, α_2, and α_3 such that

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

This leads to the system of equations

$$\begin{aligned} \alpha_1 + \alpha_2 + \alpha_3 &= a \\ \alpha_1 + \alpha_2 &= b \\ \alpha_1 &= c \end{aligned}$$

Since the coefficient matrix of the system is nonsingular, the system has a unique solution. (See next page)

Spanning set example 2

In fact we find that

$$\begin{bmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{bmatrix} = \begin{bmatrix} c\\ b-c\\ a-b\end{bmatrix}$$

Thus,

$$\begin{bmatrix}a\\ b\\ c\end{bmatrix} = c\begin{bmatrix}1\\ 1\\ 1\end{bmatrix} + (b-c)\begin{bmatrix}1\\ 1\\ 0\end{bmatrix} + (a-b)\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$$

so the three vectors span R^3.

Spanning set example 3

- Is $\{ [1, 0, 1]^T, [0, 1, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

Spanning set example 3

- Is $\{ [1, 0, 1]^T, [0, 1, 0]^T\}$ a spanning set of $\mathbb{R}^3$?

Solution:

(In this example we can clearly see the span of the 2 vectors, and easily find vectors NOT in their span.)

We should note that linear combinations of $[1, 0, 1]^T$ and $[0, 1, 0]^T$ produce vectors of the form $[\alpha, \beta, \alpha]^T$. Thus any vector $[a, b, c]^T$ in $\mathbb{R}^3$ where $a \neq c$ would not be in the span of these two vectors.

Therefore $\{[1, 0, 1]^T, [0, 1, 0]^T\}$ is NOT a spanning set of $\mathbb{R}^3$.

Spanning set example 4

- Is $\{ (1, 2, 4)^T, (2, 1, 3)^T, (4, -1, 1)^T \}$ a spanning set of $\mathbb{R}^3$?

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Spanning set example 4

- Is $\{ (1, 2, 4)^T, (2, 1, 3)^T, (4, -1, 1)^T \}$ a spanning set of $\mathbb{R}^3$?

Solution:

Similar to Example 2. If $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + \alpha_3 \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix}$ then

$$\begin{aligned} \alpha_1 + 2\alpha_2 + 4\alpha_3 &= a \\ 2\alpha_1 + \alpha_2 - \alpha_3 &= b \\ 4\alpha_1 + 3\alpha_2 + \alpha_3 &= c \end{aligned}$$

In this case, however, the coefficient matrix is singular. Gaussian elimination will yield a system of the form

$$\begin{aligned} \alpha_1 + 2\alpha_2 + 4\alpha_3 &= a \\ \alpha_2 + 3\alpha_3 &= \frac{2a - b}{3} \\ 0 &= 2a - 3c + 5b \end{aligned}$$

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If $2a - 3c + 5b \neq 0$, then the system is inconsistent. Hence, for most choices of *a*, *b*, and *c*, it is impossible to express $[a, b, c]^T$ as a linear combination of $[1, 2, 4]^T$, $[2, 1, 3]^T$, and $[4, -1, 1]^T$. The vectors do not span R^3.

Row space & column space

- For an $m \times n$ matrix A, the rows are n-vectors from $\mathbb{R}^{1\times n}$, and the column vectors are m-vectors from $\mathbb{R}^m$.

Definition

For an $m \times n$ matrix A, the subspace of $\mathbb{R}^{1\times n}$ spanned by the row vectors of A is called the **row space** of A. The subspace of $\mathbb{R}^m$ spanned by the column vectors of A is called the **column space** of A.

矩阵A的行空间：矩阵A所有行向量的张成（是$\mathbb{R}^{1\times n}$的子空间）
矩阵A的列空间：矩阵A所有列向量的张成（是$\mathbb{R}^m$ 的子空间）

Example

- The 3×4 matrix

$$A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & 0 & -4 & 3 \\ 1 & -3 & -14 & 9 \end{bmatrix}$$

has a row space in $\mathbb{R}^{1\times 4}$ given by

$$Span(\,[1,1,2,-1],[2,0,-4,3],[1,-3,-14,9])$$

and a column space in $\mathbb{R}^3$ given by

$$Span(\left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ -14 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 9 \end{bmatrix}\right)$$